

Grant for Surgical Robotic Arm for Tympanoplasty

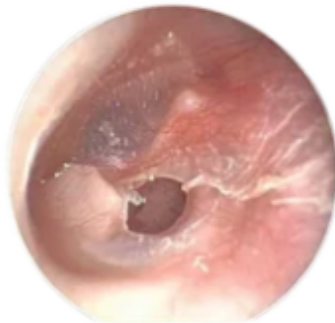
Surgical Procedure: Repairing Holes in the Eardrum Through Bio Graft Placement Through the Ear Canal

- Uses personalized pre planned surgical route.
- Inserts the graft through the ear canal, the grafts back layer sliding through the perforation, holding it in place.
- Eliminates possible hand tremors.
- Can navigate the delicate ear canal structure.
- Aligns the graft past human hand precision.
- Replaces the need for an invasive surgical route or anesthesia.
- Surgery can be performed with the patient sitting upright in any office.

Tympanic Membrane Perforations

Over 320 million people suffer from Tympanic Membrane Perforations.

Tympanic membrane perforations are holes in the eardrums thin layer of tissue.



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Caused by

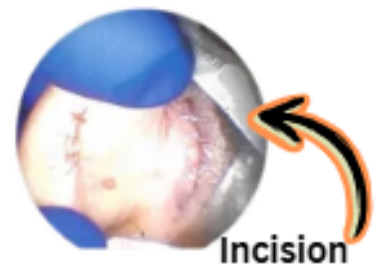
- Infections
- Air pressure changes
- Trauma
- Myringotomy procedure

Cause

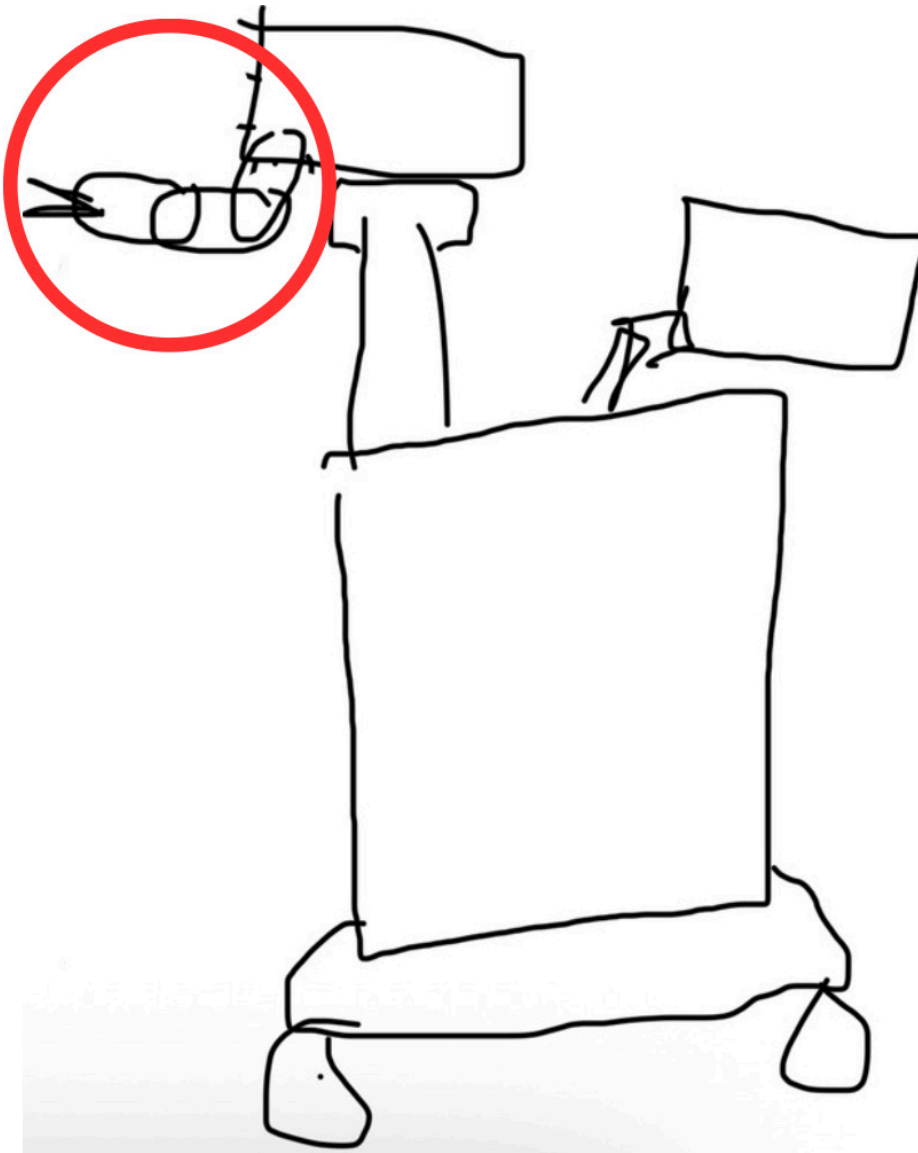
- Tinnitus
- Vertigo
- Hearing loss
- Ear Infections

Current Surgical Procedure

- Current surgical procedure expensive and inaccessible with low success rates and multiple incisions.
- Entrance to the eardrum from an incision behind the ear
- This incision is painful and takes multiple months to heal
- This access method poses the risk of hitting the hearing bones causing permanent hearing loss.
- Surgery needs anesthesia and takes multiple hours.
- Mistakes can easily occur during surgery, causing punctures of the eardrum.



Layout



The bill of materials below will cover the surgical robotic arm for graft placement and all electrical components. I will print the outer casing for the robotic arm in my robotics workshop at school which provides free filament. I will design the outer casing on solidworks which is also provided by my school. The other parts of the surgical robot including the monitor and cart, I will use my laptop and computer and purchase all other materials needed to build the cart and other accessories.

Simulation Training for the Robotic Arm

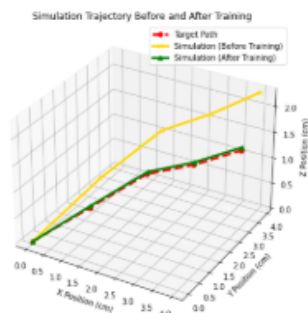
Below is my previous design for the robotic arm. The new design of the robotic arm will be created and assembled in solidworks.



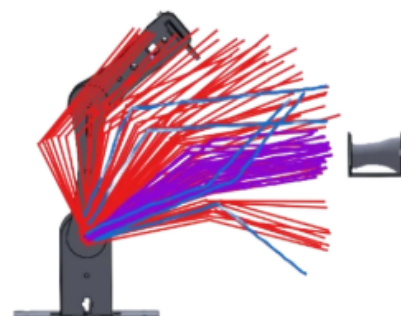
- Assembled in SOLIDWORKS assembly with screws and motors.
- Motion simulation with personalized ear looped for multiple hours.

- The simulation shows where the arm gets stuck, when it touches the canal, and how much each motor must rotate, changing these values and learning from itself.
- The simulation gives XYZ points, joint angles, and sequence of movement used to create a motion path by converting angles to servo commands and putting this information into the code.

Data Analysis



Simulation testing was preformed and each path taken during testing is mapped out to show initial errors, the learning curve, and the best trajectory. After simulation training with the personalized ear, the robot aligned with the ear canal and learned path motion for the surgery.



Previous Design I Want to Advance



Bill of Materials

Name	Amount	Link	Price
M4 x 10 mm socket-head cap screws	20	Open link	\$7.99 / pack
M3 x 8 mm low-profile button-head screws	12	Open link	\$6.99 / pack
M3 x 8 mm socket-head cap screws	50	Open link	\$6.99 / pack
M3 x 14 mm socket-head cap screws	32	Open link	\$7.99 / pack
M3 hex nuts	3	Open link	\$5.99 / pack

M3 x 25 mm socket-head cap screws	7	Open link	\$6.99 / pack
M2 x 10 mm socket-head cap screws	10	Open link	\$6.99 / pack
M3 x 16 mm socket-head cap screws	10	Open link	\$6.99 / pack
M3 x 6 mm socket-head cap screw	1	Open link	\$5.99 / pack
M3 x 12 mm socket-head cap screws	23	Open link	\$6.99 / pack
M3 x 35 mm socket-head cap screws	7	Open link	\$7.99 / pack
M4 x 16 mm socket-head cap screws	30	Open link	\$7.99 / pack
M4 x 14 mm socket-head cap screws	5	Open link	\$7.99 / pack
M4 x 50 mm socket-head cap screws	4	Open link	\$7.99 / pack
M3 x 15/16 mm countersunk screws	4	Open link	\$6.99 / pack
M5 x 16 mm socket-head cap screws	6	Open link	\$7.99 / pack
HTD3M-396 closed-loop timing belt, 6 mm wide	1	Open link	\$9.99
HTD3M-342 closed-loop timing belt, 6 mm	1	Open link	\$8.99

wide			
HTD3M-201 closed-loop timing belt, 6 mm wide	1	Open link	\$8.99
HTD3M-246 closed-loop timing belt, 6 mm wide	1	Open link	\$8.99
HTD3M 12-tooth pulley, 5 mm bore	2	Open link	\$9.99 / pair
HTD3M 15-tooth pulley, 5 mm bore	1	Open link	\$8.99
8 mm shaft coupler, M4 mounting holes	3	Open link	\$12.99 / pack
AXK3552 thrust needle bearing, 35 x 52 x 4 mm	1	Open link	\$10.99
NSK HR32906J tapered roller bearing	5	Open link	\$29.99 each
NSK HR32907J tapered roller bearing	4	Open link	\$34.99 each
3 x 8 x 4 mm ball bearing	2	Open link	\$7.99 / pack
3 x 10 x 4 mm ball bearing	6	Open link	\$8.99 / pack
12 mm illuminated on/off push button, 3-6 V	1	Open link	\$7.99
JST-PH 2.0 mm 4-pin cable/connector	1	Open link	\$7.99 / kit

24 V cooling fan	1	Open link	\$8.99
GX16 2-pin male/female power connector	1	Open link	\$8.99 / pair
M8 female 4-pin sensor connector	1	Open link	\$9.99
ZW12-3 limit switch	3	Open link	\$9.99 / pack
M4 NPN normally-open inductive sensor	1	Open link	\$13.99
GX-F8A inductive proximity sensor	1	Open link	\$16.99
M5 NPN normally-open inductive sensor	1	Open link	\$13.99
19 mm emergency-stop button	1	Open link	\$12.99
PM4 pneumatic push-in fitting	4	Open link	\$9.99 / pack
4 x 2.5 mm pneumatic tubing, 2.5 m	1	Open link	\$9.99
XT30 female connector	1	Open link	\$7.99 / pack
5/2 pneumatic solenoid valve, 24 V DC	1	Open link	\$18.99
Solenoid valve connector with silencer	2	Open link	\$11.99 / pack
PC4-M5 pneumatic	3	Open link	\$7.99 / pack

fitting			
4 mm pneumatic gripper fitting	2	Open link	\$7.99 / pack
Hook-up wire spools	2	Open link	\$12.99 each
Heat-shrink tubing assortment	1	Open link	\$7.99
Cable gland assortment	1	Open link	\$9.99
2-conductor power wire for XT30/GX16	1	Open link	\$12.99
NEMA 17 precision planetary gearbox, 20:1	2	Open link	\$39.99 each
NEMA 17 precision planetary gearbox, 10:1	1	Open link	\$32.99
NEMA 17 stepper motor, 16 Ncm, 42 x 42 x 20 mm	1	Open link	\$9.99
NEMA 17 stepper motor, 45 Ncm, 42 x 42 x 40 mm	3	Open link	\$11.99 each
NEMA 17 stepper motor, 65 Ncm, 42 x 42 x 60 mm	2	Open link	\$15.99 each
24 V, 5 A power supply	1	Open link	\$13.99

Final Price

With shipping and taxes, it will come out to \$1,018.99. I am requesting a grant for \$1000 and all other costs will be covered by me.